Behavioral Calibration for Mathematical Reasoning under Low-Data Reinforcement Learning

Controlled Experiments and Token-Level Evidence for 1-shot GRPO

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Capstone Project Midterm Presentation

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2. Experimental Design
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Context

Reinforcement learning with verifiable rewards (RLVR) has become a central recipe for improving mathematical reasoning in large language models [Shao et al., 2024, DeepSeek-AI et al., 2025].

Open Question

Recent evidence suggests that surprisingly small training sets, even one example, may trigger measurable reasoning gains [Wang et al., 2025b]. This raises a key question:

What does 1-shot RL actually improve?

- Does it teach new mathematical knowledge?
- Does it calibrate an already latent reasoning capability?
- Which model families and reward structures support this effect?

Main Claim

1-shot GRPO is best understood as a **behavioral calibration mechanism**: it improves the probability of expressing already available correct reasoning, rather than creating an entirely new capability from one sample.

Evaluation Strategy

- Compare base and instruction-tuned Qwen2.5-Math-1.5B models.
- Contrast 1-shot, full-data, weakly regularized, and corrupted-reward training.
- Move beyond score curves by analyzing token-level entropy, confidence, EOS behavior, and correctness-conditioned trajectories.

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Models, Data, and Algorithm

Models

- Qwen2.5-Math-1.5B
- Qwen2.5-Math-1.5B-Instruct

Data

- Training source: DeepScaleR subset [Luo et al., 2025]
- Full-data training: 1209 examples
- 1-shot training: one selected example
- Evaluation: MATH500

Training Method

- GRPO, implemented with VeRL/HybridFlow [Sheng et al., 2025]
- Verifiable final-answer reward
- Grouped rollout evaluation
- Token diagnostics collected from rollout traces and log-probability statistics

Seven Main Experimental Settings

Setting	Purpose	Interpretation
Base n=8	Main 1-shot run	Low-data effect on the base model.
Base full-data	Data-rich reference	Ceiling under the same recipe.
Base n=1	Group-size ablation	Importance of grouped rollouts.
Base weak-reg.	Regularization ablation	Sensitivity to regularization.
Base random-reward	Strict control	Reward semantics destroyed.
Instruct n=8	Model comparison	1-shot effect after instruction tuning.
Instruct full-data	Strong reference	Instruction-tuned upper reference.

Score-Level Metrics

- Validation accuracy on MATH500.
- Best score, final score, early gain, and endpoint retention.

Token-Level Metrics

- Mean response length.
- Mean token entropy and top-1 probability.
- EOS probability and tail confidence.
- Correctness-conditioned gaps between correct and incorrect responses.

Takeaway

The score curves answer *whether* training improves performance; token diagnostics help explain *how* the behavior changes.

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Main Validation Curves

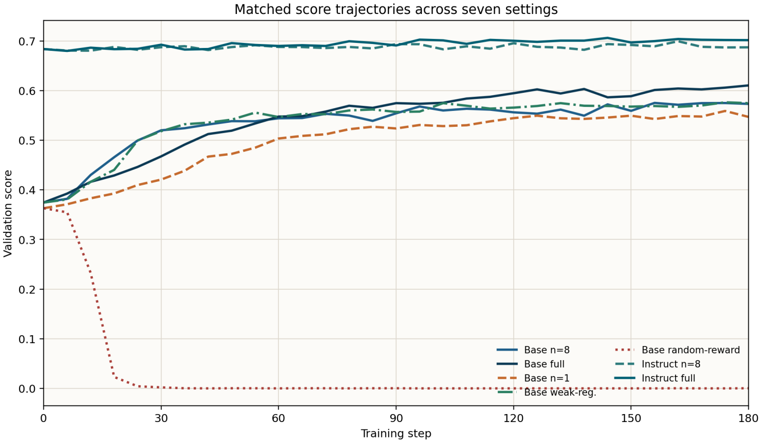

Figure 1: Validation accuracy across the seven main settings.

Empirical Pattern

- Base 1-shot training improves substantially from the initial checkpoint.
- Full-data training gives a strong reference ceiling, especially for the Instruct model.
- Random-reward training collapses, showing that reward semantics are necessary.

Interpretation

- 1-shot GRPO does not merely exploit logging artifacts.
- The gain depends on a meaningful training signal.
- The effect is stronger and more stable for the base model than for the instruction-tuned model.

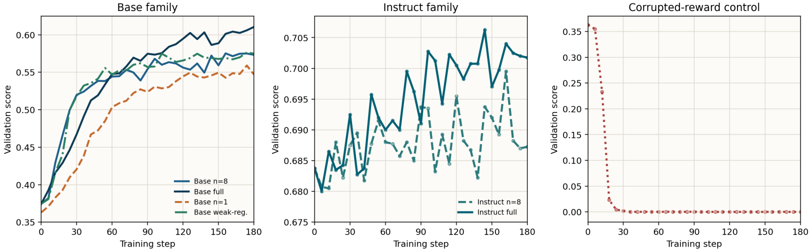

Figure 2: Zoomed score comparison excluding the collapsed control.

Takeaway

Zoomed views are necessary because the random-reward control is far outside the useful score range.

Full-Data References

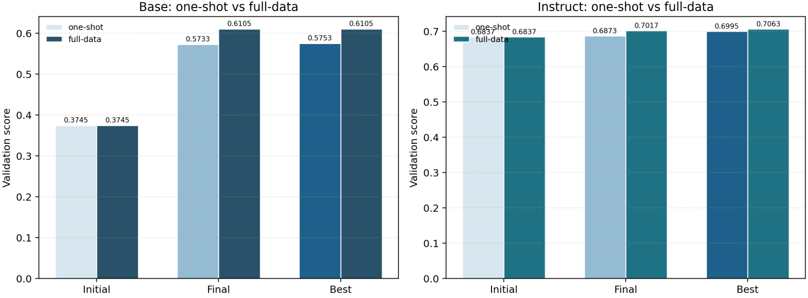

Figure 3: Matched one-shot and full-data comparisons for base and Instruct models.

Score-Level Summary

Setting	Best Acc.	Final Acc.	Role
Base n=8	0.603	0.583	Main low-data result
Base full-data	0.603	0.557	Full-data base reference
Base n=1	0.556	0.462	Group-size ablation
Base weak-reg.	0.587	0.581	Regularization ablation
Base random-reward	0.036	0.002	Strict negative control
Instruct n=8	0.095	0.079	1-shot Instruct result
Instruct full-data	0.095	0.090	Full-data Instruct reference

Takeaway

Base 1-shot reaches the same observed best accuracy as the base full-data reference; corrupted rewards fail completely.

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Motivation

Final accuracy alone cannot distinguish between several mechanisms:

- Shorter but more confident responses.
- Better calibrated termination behavior.
- Reduced uncertainty on correct trajectories.
- Collapse into overconfident but wrong outputs.

Diagnostic Principle

If 1-shot GRPO is behavioral calibration, successful runs should improve confidence and tail stability without resembling the corrupted-reward collapse.

Figure 4: Length, entropy, top-1 probability, EOS probability, and tail confidence for the seven settings.

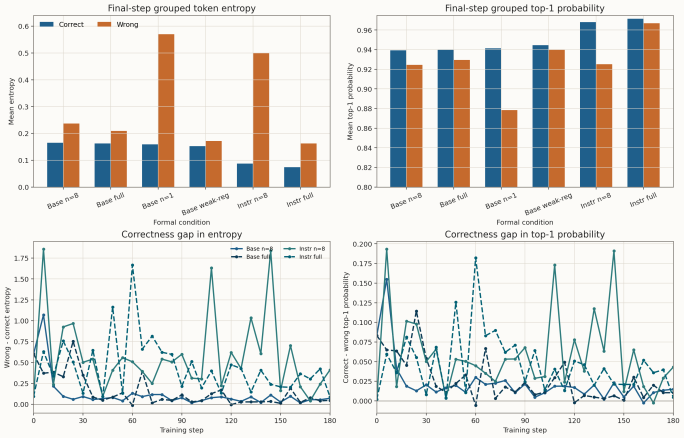

Figure 5: Correctness-conditioned token statistics and matched gap curves.

Correctness-Conditioned Findings

Setting	Correct/Wrong	Entropy C/W	Top-1 C/W	Tail C/W
Base n=8	47/17	0.165/0.237	0.939/0.925	0.957/0.866
Base full-data	51/13	0.162/0.209	0.940/0.929	0.944/0.885
Base n=1	45/19	0.159/0.570	0.941/0.878	0.949/0.785
Base weak-reg.	51/13	0.152/0.172	0.945/0.940	0.958/0.889
Instruct n=8	57/7	0.087/0.499	0.968/0.925	0.973/0.830
Instruct full-data	57/7	0.074/0.163	0.972/0.967	0.978/0.877

Stable Runs

Correct responses generally show lower entropy, higher top-1 probability, and stronger tail confidence than incorrect responses.

Full-Data Contrast

Full-data training reduces the correct/wrong confidence gap, especially for the Instruct model. This suggests a more globally calibrated distribution rather than isolated memorization of a one-example signal.

Negative Control

The random-reward setting is not a weaker version of the clean baseline. It produces a qualitatively different and collapsed behavior pattern.

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When Does the Gain Emerge?

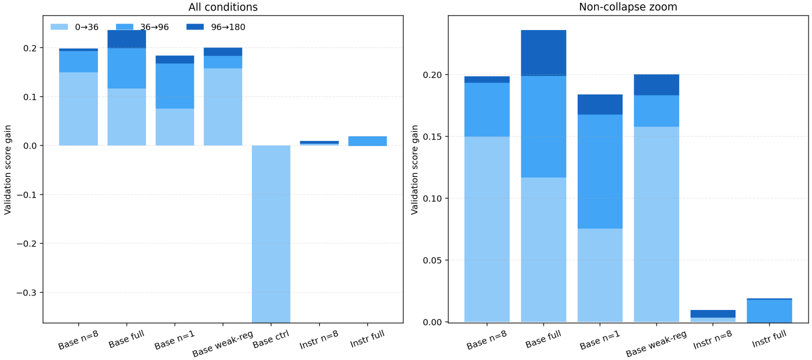

Figure 6: Early and late gain decomposition across the seven settings.

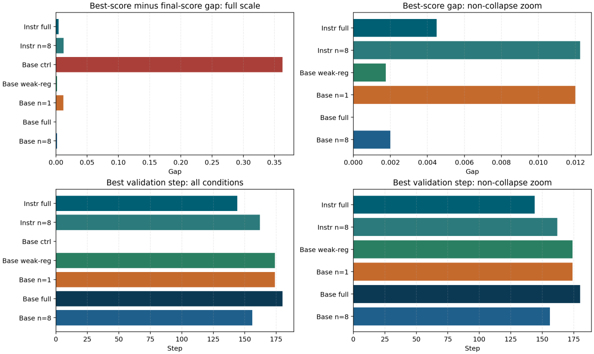

Figure 7: Gap between the best checkpoint and the final checkpoint.

Takeaway

A useful low-data method should not only reach a high peak; it should also retain performance near the endpoint.

Endpoint Frontier

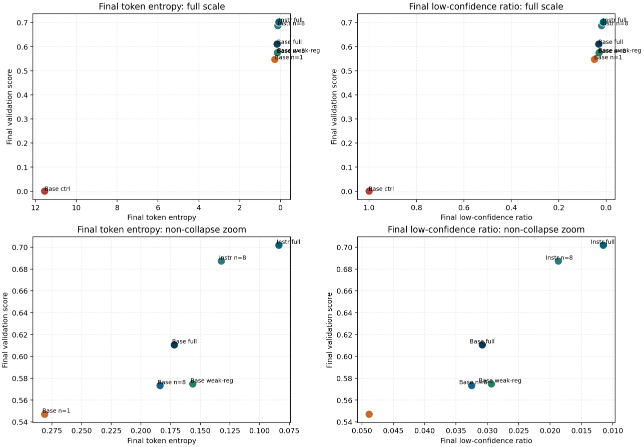

Figure 8: Endpoint behavior: performance retention and calibration-related statistics.

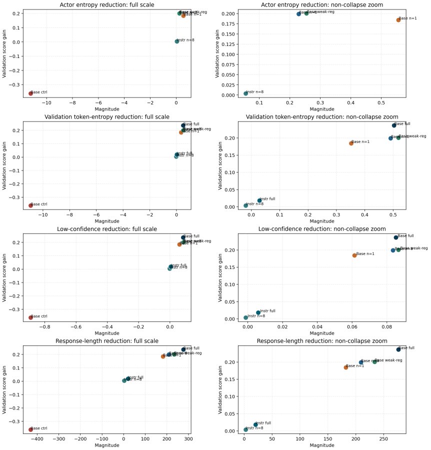

Figure 9: Relationship between score gain and token-level calibration indicators.

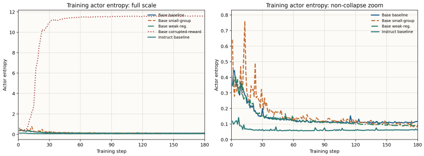

Figure 10: Actor entropy trajectories.

Observation

- Successful runs reduce uncertainty but do not resemble the collapsed random-reward behavior.
- Weak regularization remains competitive, suggesting that the useful signal is not simply stronger entropy suppression.
- This is consistent with recent analyses of entropy and high-impact tokens in RL reasoning [Cui et al., 2025, Wang et al., 2025a].

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Main Conclusions

1. **1-shot GRPO is effective for the base model.** It reaches a strong best score and remains competitive with full-data training in the observed run.
2. **The effect depends on meaningful reward semantics.** Random-reward training collapses and cannot reproduce clean-baseline gains.
3. **The mechanism is behavioral calibration.** Token-level evidence points to changes in confidence, tail stability, and correctness-conditioned uncertainty.
4. **Full-data experiments sharpen the interpretation.** They serve as reference ceilings and show what broader data coverage changes beyond one-shot learning.

Current Limitations

- Results are based on Qwen2.5-Math-1.5B; larger scales remain to be verified.
- Token diagnostics are richer than score curves but still depend on retained rollout traces and logging coverage.
- Full-data references help bound the effect, but additional seeds would strengthen statistical confidence.
- The current study focuses on mathematical reasoning; transfer to code or scientific QA is future work.

Next Steps

Near-Term

- Extend token-level tracing to more checkpoints and additional seeds.
- Compare additional low-data examples to separate sample identity from training-set size.
- Run targeted validation on EOS probability and termination behavior.

Long-Term

- Scale to larger Qwen-Math models.
- Test whether behavioral calibration transfers to code and scientific reasoning tasks.
- Develop reward/regularization designs that preserve exploration without suppressing useful reasoning paths.

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Thank You

Questions and feedback are welcome.

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